

Edu Genie-AI

AI-Powered Learning
Assistant

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Project Domain: Generative AI & Education

Project Description:

Edu Genie-AI is an intelligent web application that helps students learn topics, ask questions, create summaries, generate quizzes, and follow personalized learning paths using Artificial Intelligence. Education is becoming more digital and interactive.

- Students often need quick explanations and study materials. Traditional learning methods may not provide personalized

support for every student.

EduGenie-AI uses Generative AI to provide instant learning

- assistance.
- It acts as a digital study companion for students.

Main Purpose

• To make learning simple, interactive, personalized, and accessible using Artificial Intelligence.

Problem Statement

Students may face difficulties such as:

- Understanding complex topics.
- Finding simple explanations.
- Preparing quick study notes.
- Creating practice questions.
- Planning what to learn step by step.

Objectives

Provide instant answers to
student questions.

Explain difficult topics in simple language.

Generate short and useful summaries.

Create topic-based quizzes.

Provide structured learning paths.

- Improve independent learning.

Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Python
- Flask / FastAPI

Artificial Intelligence

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- Google

Gemini API

- Generative AI

Configuration

.env environment variables
Local development server

Working Flow

User → Web Interface → Backend API → Gemini AI
→ Response →

User

Main Features

Key Features of EduGenie-AI

1. Ask a Question

Students can ask questions and receive AI-generated answers.

2. Explain a Topic

Complex topics are explained using simple sections.

3. Summarize

Long information is converted into short and useful notes.

4. Generate Quiz

AI creates multiple-choice questions for

practice.

5. Learning Path

Creates a step-by-step roadmap for learning a topic

Explain Topic & Summarize

Learning Content Generation

Explain a Topic

The system provides:

Definition – Basic meaning of the topic.

How It Works – Main working concepts.

- Example – Simple practical example.
- Key Points – Important points for revision.

Summarize

The system:

- - Understands the given topic.
 - Identifies important information.
 - Removes unnecessary details.
 - Produces a short and easy-to-understand summary.
- This helps students learn and revise quickly.

Slide 7 – Quiz & Learning Path

Quiz Generation

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- EduGenie-AI automatically creates quizzes based on the selected

-
- topic.
- Multiple-choice questions

Four options for each question

Correct answer provided

Unique questions

Unique answer options

Useful for self-assessment

Learning Path

The system creates a structured learning roadmap:

• Basics → Core Concepts → Practice → Advanced Topics →

Project

This helps students learn systematically.

System Working & Error Handling

How the System Works

Step 1: Student enters a topic or question.

Step 2: Frontend sends the request to the backend.

Step 3: Backend processes the request.

Step 4: The request is sent to the Generative AI model.

Step 5: AI generates the required content.

Step 6: Backend validates the response.

Step 7: The result is displayed on the webpage.

Error Handling

Input validation

API error handling

Response
validation

- Demo Mode / fallback mechanism
- Clear error messages

Advantages & Future Scope

Advantages

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- Easy to use
-
- Fast responses
-
- Personalized learning support
- Useful for different subjects
- Helps with revision
-
- Generates interactive quizzes

Provides structured learning guidance

Future Scope

-
- User login and authentication
-
- Student progress tracking
- Personalized dashboards
- PDF export
-
- Multi-language support
-
- Voice-based learning
-
- More advanced learning recommendations

Conclusion

EduGenie-AI is an AI-powered educational assistant designed to

support students in their learning journey.

It combines:

- Artificial Intelligence

- Web Technology

- Interactive Learning

- Automated Content Generation

The project helps students understand topics, get answers,

revise information, practice through quizzes, and follow

structured learning paths.

Thank You